# Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A survey was conducted on students' preferred modes of transportation to school. The results are summarized in the two-way frequency table below. \\ | | Bus | Bicycle | Walking | Total | |----------------|-----|---------|---------|-------| | Under 16 | 45 | 30 | 25 | 100 | | 16 and Over | 60 | 15 | 25 | 100 | | Total | 105 | 45 | 50 | 200 | Let B be the event that a randomly selected student prefers to travel by Bus. Let U be the event that a randomly selected student is Under 16 years old. \\ What is the probability that a student prefers to travel by Bus, given that they are Under 16 years old? Mathematically, this is denoted as $P(B|U)$.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A civil engineer measures the width of a river $BC$ by setting up parallel stakes such that $DE \parallel BC$. Point $D$ is on line $AB$ and $E$ is on line $AC$. If $AD = 45$ m, $DB = 75$ m, and $DE = 36$ m, what is the width of the river $BC$?

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A 5.5-foot-tall surveyor casts a 4.4-foot shadow on level ground. At the exact same moment, a nearby radio antenna tower casts a 36-foot shadow. How tall is the antenna tower?

**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In $\Delta ABC$, segment $DE \parallel BC$. The segment lengths are $AD = x$, $DB = x + 3$, $AE = x + 2$, and $EC = x + 7$. What is the value of $x$?

**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In $\Delta ABC$, ray $AD$ bisects $\angle BAC$ with $D$ lying on side $BC$. If $AB = x + 3$, $AC = 2x$, $BD = 4$, and $DC = 6$, what is the value of $x$?

**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an hourglass figure where $AB \parallel CD$ with intersection at $E$, $AE = 2x + 1$, $EC = 5x - 4$, $BE = 6$, and $ED = 9$. What is the value of $x$?

**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In right $\Delta ABC$ with altitude $CD \perp AB$, $AD = 2$ and $DB = 8$. What is the ratio of the legs $\frac{AC}{BC}$?

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two similar triangles have areas of $45\ \text{cm}^2$ and $80\ \text{cm}^2$. If the perimeter of the smaller triangle is $27\ \text{cm}$, what is the perimeter of the larger triangle?

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

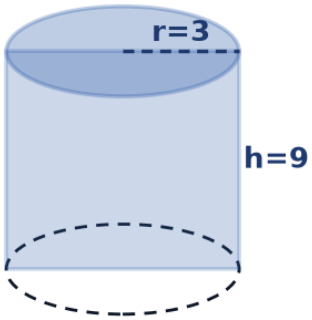
The point P has coordinates $(-3, 4)$. Point P' is the image of P after a reflection across the line $y = x$. What are the coordinates of P'?

**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right $\Delta ABC$ with right angle at $C$ and altitude $CD \perp AB$, leg $BC = 12$ and hypotenuse $AB = 16$. What is the length of segment $DB$?

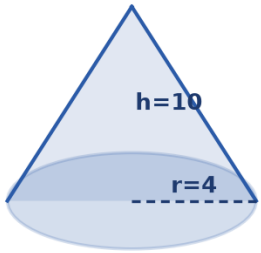
**11** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(5)$.




**12** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(2)$.




**13** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

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**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters $k$. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A circular archery target has three concentric rings of radii $10\text{ cm}$, $20\text{ cm}$, and $30\text{ cm}$. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii $10\text{ cm}$ and $20\text{ cm}$)?

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles

A circle is inscribed inside a square of side length $10\text{ cm}$. A point is chosen at random inside the square. What is the exact probability that the point lands outside the circle (in the four shaded corners)?

**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume

[Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

A cone and a cylinder have identical base radii of $6\text{ cm}$. If both solids have the same volume of $180\pi\text{ cm}^3$, find the height of the cylinder and the height of the cone.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A rhombus has diagonals of lengths $16\text{ cm}$ and $12\text{ cm}$. Find the perimeter and area of the rhombus.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

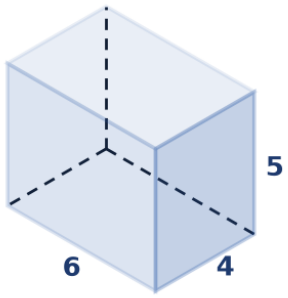
From an external point $P$, two tangents $PA$ and $PB$ are drawn to a circle with radius $r = 5\text{ cm}$. The distance from $P$ to the center of the circle $O$ is $13\text{ cm}$. Find the length of the tangents $PA$ and $PB$, and find the perimeter of quadrilateral $PAOB$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(2)$.




**a.** 12

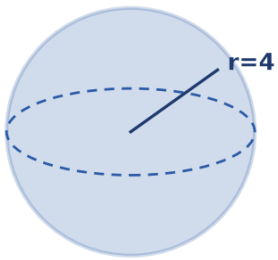
**b.** 15

**c.** 6

**d.** 16

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(2)$.




**a.** 15

**b.** 19

**c.** 8

**d.** 19

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cross sections of solids consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solids of revolution consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of prisms and cylinders consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5

**d.** 23.5

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of spheres and hemispheres consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 3$ in volume of spheres and hemispheres, evaluate $f(3)$.

**a.** 9

**b.** 11

**c.** 6

**d.** 13

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**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of composite solids consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 4$ in volume of composite solids, evaluate $f(3)$.

**a.** 13

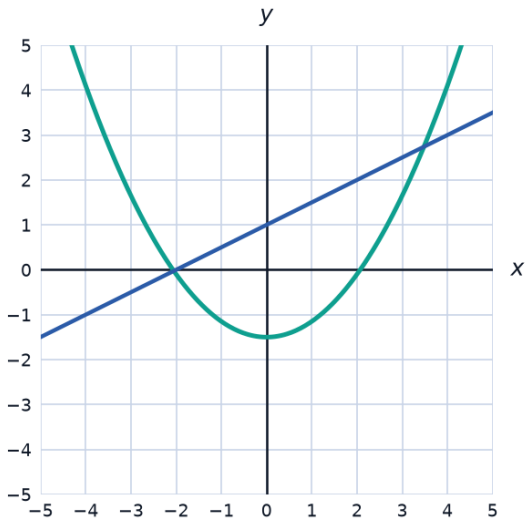
**b.** 16

**c.** 9

**d.** 17

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A point is chosen at random on a segment of length 11 cm. Find the probability it lands within 2 cm of the left endpoint.




**a.** $\frac{1}{6}$

**b.** $\frac{2}{11}$

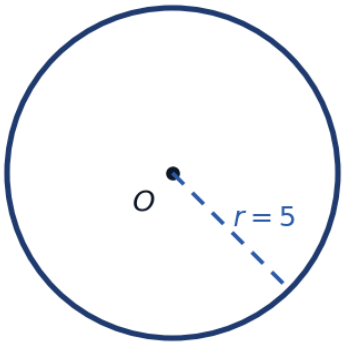
**c.** 1

**d.** $\frac{9}{11}$

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**31** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A circular dartboard has radius 15 cm with a bullseye of radius 1 cm at the center. Assuming a dart lands at a uniformly random point on the board, find the probability it lands in the bullseye.




**a.** $\frac{1}{226}$

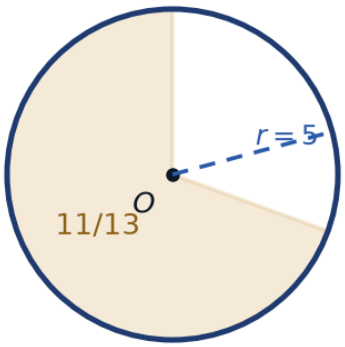
**b.** $\frac{1}{225}$

**c.** $\frac{224}{225}$

**d.** $\frac{1}{15}$

**32** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The probability that an event occurs is $\frac{11}{13}$. What is the probability that the event does NOT occur?




**a.** 0

**b.** $\frac{2}{13}$

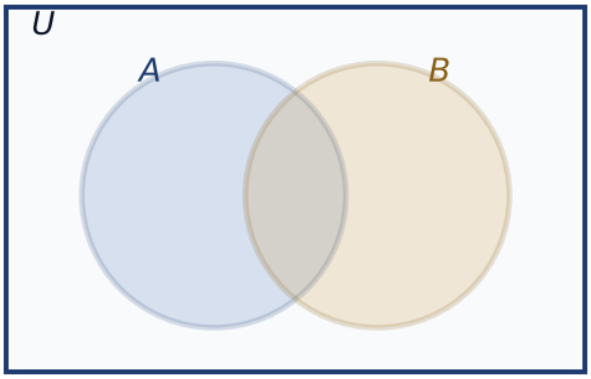
**c.** 1

**d.** $\frac{11}{13}$

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**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{2}{5}$. Find $P(A \text{ or } B)$.




**a.** $\frac{3}{5}$

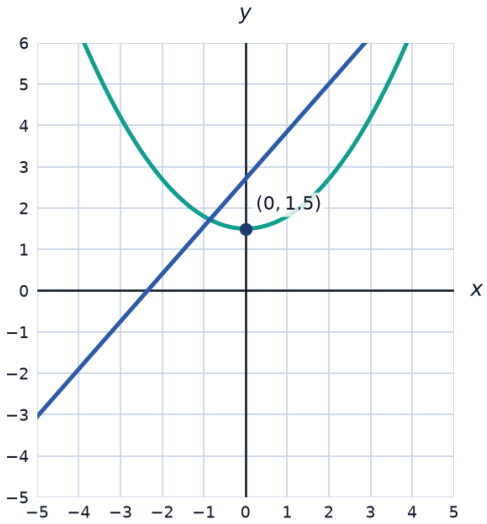
**b.** $\frac{14}{25}$

**c.** $\frac{2}{25}$

**d.** $\frac{13}{25}$

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circular target of radius 11 cm has a ring-shaped region between radius 3 cm and radius 9 cm. Find the probability a randomly landing dart hits this ring.




**a.** $\frac{9}{121}$

**b.** $\frac{81}{121}$

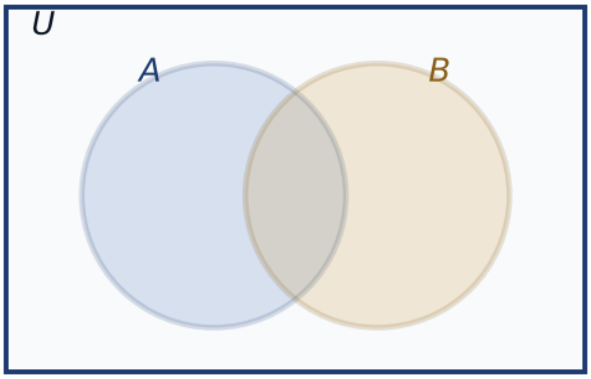
**c.** $\frac{72}{121}$

**d.** $\frac{36}{61}$

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**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{6}{12}$. Find $P(A \text{ and } B)$.




**a.** $\frac{1}{10}$

**b.** $\frac{2}{11}$

**c.** $\frac{1}{5}$

**d.** $\frac{1}{2}$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A quality control laboratory receives components sorted into two bins: - **Bin A** contains 10 components, of which exactly 3 are defective and 7 are non-defective. - **Bin B** contains 10 components, of which exactly 6 are defective and 4 are non-defective. To conduct an inspection, an automated system first selects a bin at random: it rolls a fair standard six-sided die and selects **Bin A** if the result is 1 or 2; otherwise (if the result is 3, 4, 5, or 6), it selects **Bin B**. From the chosen bin, the system then randomly samples 2 components without replacement. If both sampled components are found to be defective, what is the probability that the components were drawn from **Bin A**?

**a.** $\frac{1}{11}$

**b.** $\frac{1}{9}$

**c.** $\frac{1}{6}$

**d.** $\frac{10}{11}$

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A random sample of 200 high school students was surveyed regarding their grade level and whether they participate in school sports. The results are summarized in the two-way frequency table below: | | Participates in Sports ($S$) | Does Not Participate in Sports | Total | | :--- | :---: | :---: | :---: | | Grade 10 ($T$) | 48 | 32 | 80 | | Grade 11 | 42 | 78 | 120 | | Total | 90 | 110 | 200 | A student is selected at random from this sample. Let $S$ be the event that the student participates in sports, and let $T$ be the event that the student is in Grade 10. Which of the following statements is correct regarding the conditional probability $P(S \mid T)$ and the statistical independence of events $S$ and $T$?

**a.** $P(S \mid T) = 0.60$, and the events are not independent because $P(S \mid T) eq P(S)$.

**b.** $P(S \mid T) = 0.60$, and the events are independent because $P(S \mid T) > P(S)$.

**c.** $P(S \mid T) = 0.24$, and the events are not independent because $P(S \mid T) eq P(S)$.

**d.** $P(S \mid T) = 0.40$, and the events are independent because $P(S \mid T) = P(T)$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

A bag contains 5 red marbles and 4 blue marbles. Two marbles are drawn from the bag without replacement. What is the probability that the first marble drawn is red and the second marble drawn is blue?

**a.** $\frac{5}{18}$

**b.** $\frac{4}{18}$

**c.** $\frac{20}{81}$

**d.** $\frac{5}{9}$

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A point is chosen uniformly at random on the line segment from $x = 0$ to $x = 20$. Let event $A$ be the event that the point lands on the interval $[2, 10]$, and let event $B$ be the event that the point lands on the interval $[6, 16]$. What is the probability that the point lands in $A$ or $B$, $P(A \cup B)$?

**a.** $\frac{7}{10}$

**b.** $\frac{9}{10}$

**c.** $\frac{1}{5}$

**d.** $\frac{1}{2}$

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**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

A bag contains 18 marbles, of which 12 are red. If one marble is drawn at random, what is the probability it is red?

**a.** $\frac{1}{3}$

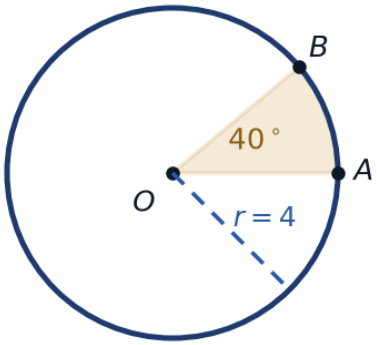
**b.** $\frac{2}{3}$

**c.** $\frac{3}{2}$

**d.** $\frac{12}{19}$

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Lines & Angles

Two secants are drawn from an external point to a circle, intercepting arcs of $100^\circ$ and $40^\circ$. What is the measure of the angle formed by the secants?




**a.** 100°

**b.** 30°

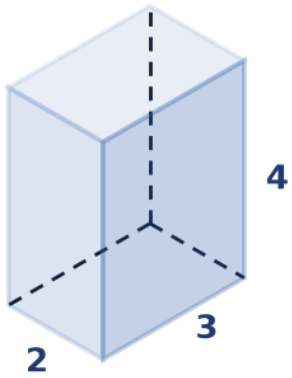
**c.** 70°

**d.** 40°

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**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Triangles & Congruence

Volume of a rectangular prism with dimensions $2 \times 3 \times 4$:




**a.** 21

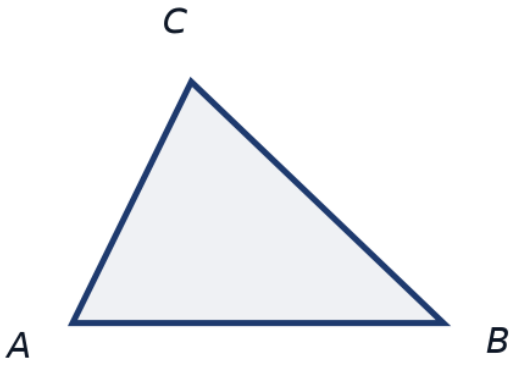
**b.** 9

**c.** 24

**d.** 26

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In a triangle, the two remote interior angles measure $14^\circ$ and $10^\circ$. What is the measure of the exterior angle at the third vertex?




**a.** 24°

**b.** 14°

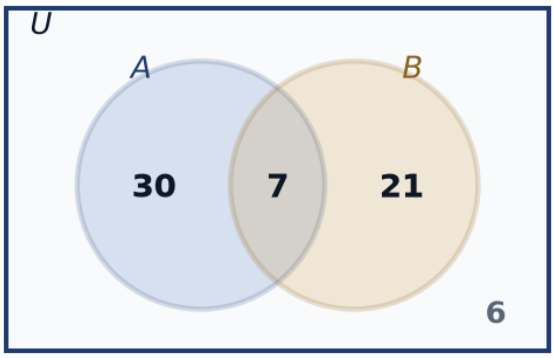
**c.** 156°

**d.** 10°

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**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Two chords intersect inside a circle, intercepting arcs of $30^\circ$ and $178^\circ$ (the arc and its vertical arc). What is the measure of the angle formed at the intersection?




**a.** 76°

**b.** 208°

**c.** 30°

**d.** 104°

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A rectangle has length 11 and width 6. What is its area?



**a.** 34

**b.** 66

**c.** 72

**d.** 17

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